SISTEM PERSAMAAN

Tentukan solusi (x, y) dari sistem persamaan dibawah ini!

$x - 2(3) = -4$ | $x - 6 + 6 = -4 + 4$
$x - 6 = -4$ | $x = -2$

I. Tentukan solusi dari SPLDV berikut dengan metode substitusi! $x = -2$ $y = 3$

a. $\begin{cases} 2x + 3y = 13 \\ x - 2y = -4 \end{cases}$ $\begin{vmatrix} 1 \\ 2 \end{vmatrix}$ $2x + 3y = 13$, $2x - 4y = -8$ −, $7y = 21$, $y = 3$

$x = 2$ $y = 3$

b. $\begin{cases} 2x + 3y = 13 \\ 4x - y = 5 \end{cases}$ $4x - 3 = 5$, $4x - 3 + 3 = 5 + 3$, $4x = 8$, $x = 2$, $y = 3$

$x = 25$ $y = -3$

c. $\begin{cases} x + 4y = 13 \\ 2x - y = 53 \end{cases}$ $\begin{vmatrix} 2 \\ 1 \end{vmatrix}$ $2x + 4y = 26$, $2x - y = 53$, $x = 25$

d. $\begin{cases} 4x + y = -14 \\ x - 2y = 1 \end{cases}$ $4x + (-2) = -14$, $4x = -12$, $x = -3$ $y = -2$

$x = -3$
$y = -2$

$x = 2$, $y = 2$, $y = 2.5$, $x = 0$, $y = 1$ $x = 2$, $y = 3$, $y = 1$

c. $\begin{cases} 5x - 3y = 16 \\ 3x - 4y = 14 \end{cases}$ $\begin{vmatrix} 4 \\ 3 \end{vmatrix}$ $20x - 12y = 64$, $9x - 12y = 42$

d. $\begin{cases} 4x + 2y = 5 \\ 2x + 3y = 7{,}5 \end{cases}$ $4x + 2(2.5) = 5$, $4x + 5 = 5$, x:

e. $\begin{cases} 2x + 3y = 7 \\ 2x - 4y = 14 \end{cases}$ $2x + 3y = 7$, $2x - 4y = 14$, $7y = 7$, $y = 1$

f. $\begin{cases} 2x + 5y = 1 \\ 2x + y = 5 \end{cases}$

II. Tentukan solusi dari SPLDV berikut dengan metode eliminasi!

a. $\begin{cases} 4x - 5y = 7 \\ 2x + 3y = 4 \end{cases}$ $\begin{vmatrix} 3 \\ 5 \end{vmatrix}$ $12x - 15y = 21$, $10x + 15y = 20$ +, $y = \frac{1}{11}$

$y = \frac{1}{11}$ $x = \frac{41}{22}$

b. $\begin{cases} 3x - y = 4 \\ 2x + 2y = 8 \end{cases}$ $\begin{vmatrix} 2 \\ 1 \end{vmatrix}$ $6x - 2y = 8$, $2x + 2y = 8$ +, $8x = 16$

$x = 2$ $y = 2$

c. $\begin{cases} 2x + 3y = 13 \\ 4x - y = 5 \end{cases}$ $\begin{vmatrix} 1 \\ 3 \end{vmatrix}$ $2x + 3y = 13$, $12x - 3y = 15$ +, $y = 3$, $x = 1$

$x = 3$ $y = 1$

d. $\begin{cases} 3x + 2y = 12 \\ 4x - 3y = 1 \end{cases}$ $\begin{vmatrix} 3 \\ 2 \end{vmatrix}$ $9x + 6y = 36$, $8x - 6y = 2$ +, $17x = 38$, $x = \frac{38}{17}$, $y = 3$

e. $\begin{cases} 4x + 6y = 34 \\ 2x + 6y = 26 \end{cases}$

III. Tentukan solusi dari SPLDV berikut dengan metode campuran!

$3x + 2(3) = 12$
$3x + 6 - 6 = 12 - 6$
$3x = 6$
$x = 2$

a. $\begin{cases} 3x + 2y = 12 \\ 4x + 3y = 17 \end{cases}$ $\begin{vmatrix} 4 \\ 3 \end{vmatrix}$ $12x + 8y = 48$, $12x + 9y = 51$

$x = 2$ $y = 3$

b. $\begin{cases} 4x - 2y = 2 \\ 5x + 4y = 35 \end{cases}$ $\begin{vmatrix} 2 \\ 1 \end{vmatrix}$ $8x - 4y = 4$, $5x + 4y = 35$ +, $13x = 39$, $x = 3$

$x = 3$ $y = 5$

$4(3) - 2y = 2$
$12 - 2y = 2$
$12 - 12 - 2y = 2 - 12$
$y = 5$